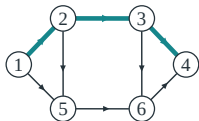


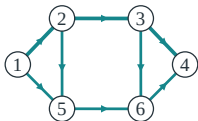
Constraint-like limit



$\varepsilon \rightarrow 0$

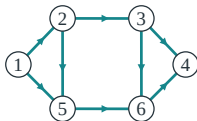
Constrained graph LP

Entropic graph solve



finite ε

Penalty-like limit



$\varepsilon \rightarrow \infty$

Clipped active-set price

Same graph and budgets; vary ε

SafeOT-Dual: one price rule

$$\lambda = \underbrace{\text{filtered clip}(\lambda_{\varepsilon}^*, 0, c)}_{\text{graph-solved price}} + \underbrace{\beta}_{\text{cost-feedback price}}$$

$$A_{\text{update}} = (A_r - \sum_k \lambda_k A_{c_k}) / (1 + \sum_k \lambda_k)$$

$c = 0$

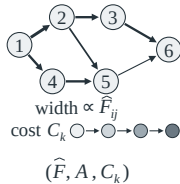
Fixed β : fixed penalty

Adaptive β : Lagrangian

$b_k = B_k / L$
per-step budgets

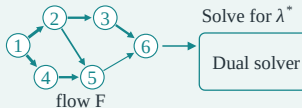
Hard edges H
zero flow

Empirical graph



GAE: A_r, A_c

Joint budget pricing



Solve for λ^*

Dual solver

$$\min_F -\langle A, F \rangle + \varepsilon \text{KL}(F \| \hat{F})$$

$$\text{Flow balance} \mid \langle C_k, F \rangle \leq b_k \mid F_H = 0$$

λ^*

Clip +
smooth

$\bar{\lambda}^*$

Priced policy update

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO / TRPO $\rightarrow \pi_{\text{next}}$

Episode costs $\bar{J}$

episode budget B

Cost feedback

$$\beta \leftarrow [\beta + \eta(\bar{J} - B)]_+$$

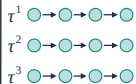
β

+

λ

next rollout

Rollout batch



next rollout